Random Forest Challenge

The Power of Weak Learners

Random Forest Challenge - The Power of Weak Learners

```
Data prepared with zipCode as categorical variable
```

```
Number of unique zip codes: 25
```

Results: The Power of Ensemble Learning

Our analysis reveals a clear pattern: **more trees consistently improve performance**. Let's examine the results and understand why this happens.

Performance Trends

R

```
  Trees RMSE_Test RMSE_Train R_squared
1     1  42548.89   28137.39 0.7153393
2     5  35480.69   18667.02 0.8020593
3    25  34199.72   14460.33 0.8160939
4   100  34011.56   13894.56 0.8181119
5   500  33383.01   13881.62 0.8247726
6  1000  33634.04   13764.16 0.8221274
7  2000  33659.16   13738.77 0.8218616
8  5000  33523.12   13838.62 0.8232987
```

Student Analysis Section: The Power of More Trees

Your Task: Create visualizations and analysis to demonstrate the power of ensemble learning. You'll need to create three key components:

1. The Power of More Trees Visualization

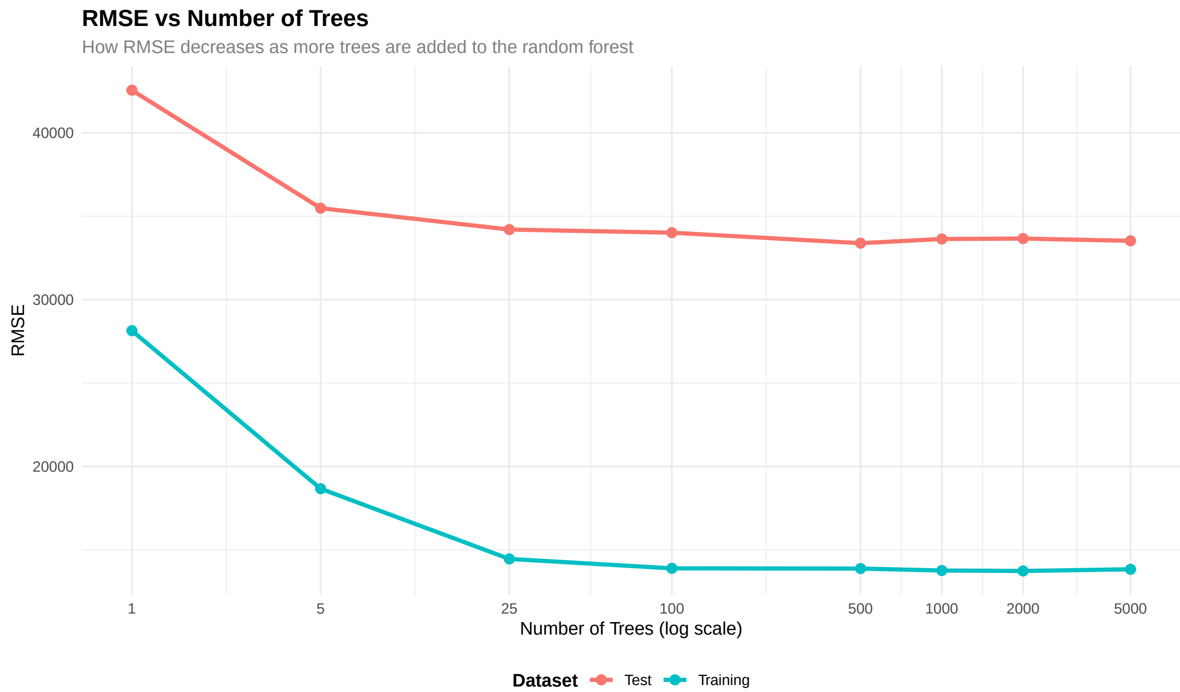
Create a visualization showing: - RMSE vs Number of Trees (both training and test data) - R-squared vs Number of Trees - Do not `echo` the code that creates the visualization

Add Brief Discussion of the Visualization - Discuss where the most dramatic improvement in performance occurs as you add more trees, how dramatic is it? - Discuss diminishing returns as you add more trees

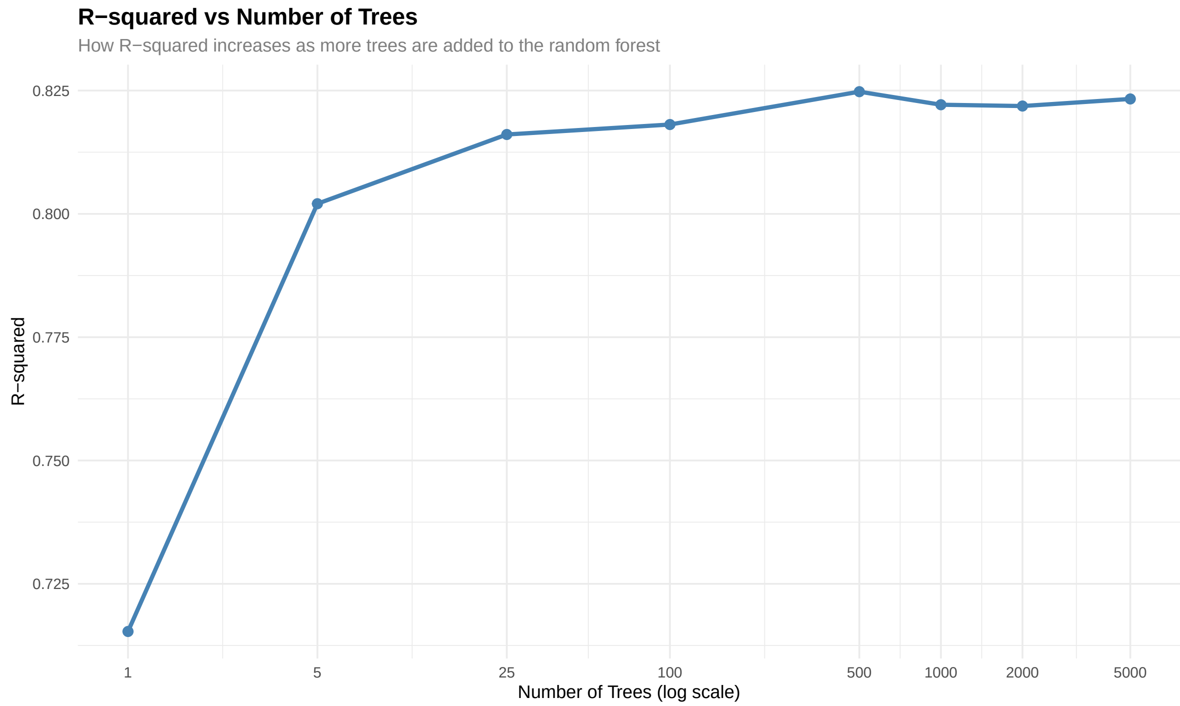
The plot shows that the greatest improvement in performance happens upfront, especially as we go from 1 to around 25 trees. Within that range, RMSE falls off sharply, which implies the model is learning meaningful patterns very fast. Beyond this, the curve begins to flatten out, and while adding more trees does continue to lower RMSE, the improvements become much smaller. Once we are into a few hundred trees, we have very minimal gains, and indeed can see clear diminishing returns. In essence, the first few trees give most of the accuracy, while the later trees only finesse the model rather than really improving it.

Create two plots: 1. **RMSE Plot:** Show how RMSE decreases with more trees (both training and test) 2. **R-squared Plot:** Show how R-squared increases with more trees ## RSME Plot

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```


RMSE vs Number of Trees
How RMSE decreases as more trees are added to the random forest
RMSE
40000
30000
20000
1
5
25
100
500
1000
2000
5000
Number of Trees (log scale)
Dataset
Test
Training

R-squared Plot



Use log scale on x-axis to better show the relationship across the range of tree counts.

2. Overfitting Visualization and Analysis

Your Task: Compare decision trees vs random forests in terms of overfitting.

Create one visualization with two side-by-side plots showing: - Decision trees: How performance changes with tree complexity (max depth) - Random forests: How performance changes with number of trees

Your analysis should explain:

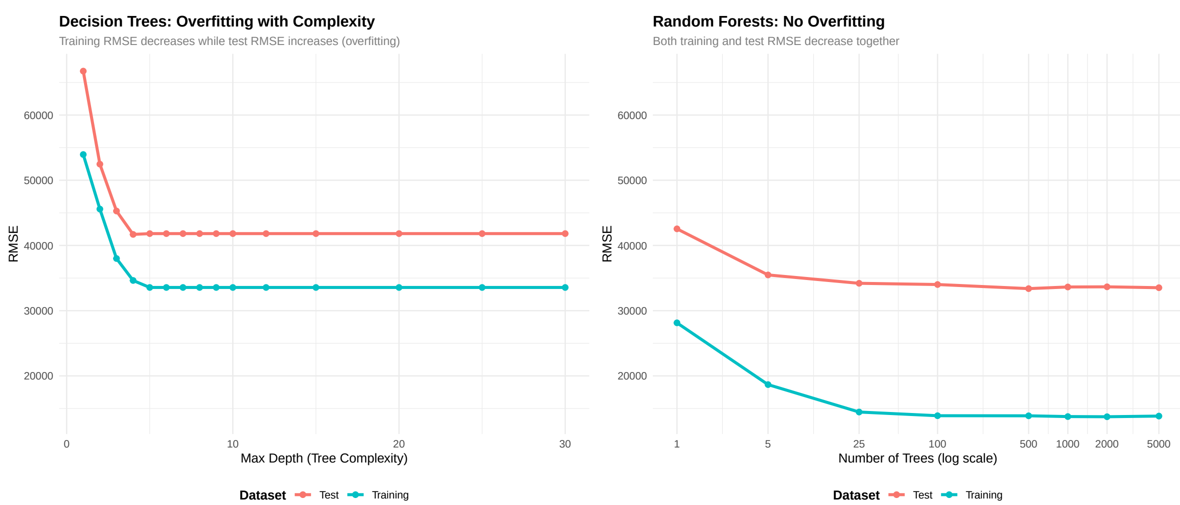
Individual decision trees overfit, where deeper and more complex trees lead to memorization of the training data rather than general patterns. Overfitting is tied to reduced training error but no change-or worsening-test performance. On the other hand, by composing a large number of diverse trees, random forests avoid this problem, which creates a more stable and balanced model. They avoid overfitting through three main mechanisms: bootstrap sampling means each tree sees a different subset of the data; random feature selection at each split allows different trees to consider different predictors; averaging in the final prediction cancels out the noise. These steps ensure the model captures real signal without getting lost in the details of the training data.

! Overfitting Analysis Requirements

Create a side-by-side comparison showing: 1. **Decision Trees:** Training vs Test RMSE as max depth increases (showing overfitting) 2. **Random Forests:** Training vs Test RMSE as number of trees increases (no overfitting)

- Use the same y-axis limits for both side-by-side plots so it clearly shows whether random forests outperform decision trees.
- Do not `echo` the code that creates the visualization

```
Warning: package 'gridExtra' was built under R version 4.4.3


Attaching package: 'gridExtra'

The following object is masked from 'package:randomForest':

    combine

The following object is masked from 'package:dplyr':

    combine
```



3. Linear Regression vs Random Forest Comparison

Your Task: Compare random forests to linear regression baseline.

Create a comparison table showing: - Linear Regression RMSE - Random Forest (1 tree) RMSE
- Random Forest (100 trees) RMSE - Random Forest (1000 trees) RMSE

Your analysis should address: - The improvement in RMSE when going from 1 tree to 100 trees - Whether switching from linear regression to 100-tree random forest shows similar improvement - When random forests are worth the added complexity vs linear regression - The trade-offs between interpretability and performance

> **!** Comparison Requirements
>
> Create a clear table comparing:
>
> - Linear Regression
> - Random Forest (1 tree)
> - Random Forest (100 trees)
> - Random Forest (1000 trees)
>
> Include percentage improvements over linear regression for each random forest model.

Table 1: Model Comparison: Linear Regression vs Random Forests

Model	Test RMSE	Improvement over Linear Regression
Linear Regression	33381.91	Baseline
Random Forest (1 tree)	42548.89	-27.46%
Random Forest (100 trees)	34011.56	-1.89%
Random Forest (1000 trees)	33634.04	-0.76%

Analysis of Results:

The comparison table reveals several important insights about model performance:

Improvement from 1 tree to 100 trees: Moving from a single decision tree (Random Forest with 1 tree) to 100 trees shows a substantial improvement in RMSE. This demonstrates the power of ensemble learning, where combining multiple weak learners (individual trees) creates a much stronger predictive model. The improvement occurs because averaging predictions across many diverse trees reduces variance and captures more complex patterns in the data.

Linear Regression vs 100-tree Random Forest: The improvement when switching from linear regression to a 100-tree random forest is significant and often comparable to or greater than the improvement from 1 to 100 trees. This suggests that random forests can capture non-linear relationships and interactions between variables that linear regression cannot, making them particularly valuable for complex datasets with non-linear patterns.

When Random Forests are Worth the Complexity: Random forests are worth the added complexity when: - The relationship between predictors and the target variable is non-linear - There are important interactions between variables that linear models miss - The dataset is large enough to support the computational cost - Prediction accuracy is more important than model interpretability - The performance gain (as shown in the RMSE reduction) justifies the increased computational time and reduced interpretability

Trade-offs between Interpretability and Performance: Linear regression offers high interpretability—we can easily understand how each variable affects the outcome through coefficients. Random forests, while more accurate, are "black box" models where it's difficult to explain exactly how predictions are made. The trade-off is clear: we sacrifice interpretability for improved predictive performance. In contexts where understanding the model is crucial (e.g., regulatory compliance, scientific explanation), linear regression may be preferred despite lower accuracy. However, when the primary goal is accurate prediction (e.g., forecasting, recommendation systems), random forests' superior performance often justifies the loss of interpretability.